SOFTWARE ENGINEERING MINI PROJECT · TEAM E15

Test Case Update

Railway Reservation System · Test Plan §5.2

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PES1UG24CS280

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Standalone role-specific copy

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Document Control

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Revision History

Version	Date	Change	Author
1.0	29 Sep 2026	Ten functional, non-functional and security case specifications.	Muhammad Affan Dumba
1.1	29 Sep 2026	Added explicit negative boundaries, hash inspection, late cancellation and booking/cancellation audit verification.	Muhammad Affan Dumba

Submission relationship. This is the separate role-specific Deliverable 4 file. Its ten cases are also incorporated into Test Plan §5.2, satisfying the instruction that the update appear in the Test Plan document.

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1. Purpose and Execution Rules

This update specifies ten cases that cover the RRS functional, non-functional and security requirements. It is controlled as Test Plan §5.2 and supplied separately to identify the Deliverable 4 owner.

- Use synthetic P1/P2 passenger accounts and A1 administrator; never use real passenger/payment data.
- Reset independent cases to a known seed. Record build, environment, tester, date, actual result, evidence and status.
- A case passes only when every expected result is observed. Redact passwords and tokens from evidence.

2. Test Cases

TC-01 — Registration, Password Boundary and Credential Storage

Preconditions	No P1 account; authorized test-only storage/log access.
Steps	Submit 11-character password; then valid 12-character password and unique email; repeat email; inspect stored credential, response and logs.
Expected	11-character and duplicate inputs are rejected; exactly one account is created; stored value is a salted adaptive hash unequal to the password; plaintext is absent from response/logs.
Traceability	FR-01; SEC-02

TC-02 — Sign-In, Lockout and Authentication Audit

Preconditions	P1 exists and is unlocked.
Steps	Sign in correctly and out; submit five wrong passwords within 15 minutes; attempt a sixth; inspect audit; retry valid credentials after 15 minutes.
Expected	Valid sign-in works; errors are generic; sixth attempt is rejected for 15 minutes; authentication outcomes are audited without passwords; valid access resumes after the period.
Traceability	FR-02, FR-09; SEC-02, SEC-04

TC-03 — Search and Availability

Preconditions	Matching, nonmatching, sold-out and multiple-departure trips seeded.
Steps	Search by origin, destination and date; inspect filtering, ordering and seat counts.
Expected	Only exact matches appear in departure order; every count remains from zero through capacity.
Traceability	FR-03, FR-04

TC-04 — Successful Booking, History and Audit

Preconditions	P1 signed in; selected trip has one free seat; simulator succeeds.
Steps	Book once; query P1 history and audit records.
Expected	One confirmed booking, unique PNR and one seat allocation; P1 history contains all required fields; confirmation audit contains actor/action/time/outcome.
Traceability	FR-05, FR-06, FR-09

TC-05 — Payment Failure and Idempotent Retry

Preconditions	P1 signed in; one free seat; known idempotency key.
Steps	Run once with simulator failure; then use success and send the same request twice with one new idempotency key.
Expected	Failure confirms nothing and releases the hold; duplicate successes produce one booking/PNR and allocate one seat.
Traceability	FR-05; NFR-03

TC-06 — Ownership, Eligible/Late Cancellation and Audit

Preconditions	P1 owns future and departed confirmed bookings; P2 signed in separately.
Steps	P2 reads/cancels P1’s future booking; P1 cancels it twice; P1 attempts departed-booking cancellation; inspect inventory/refund/audit.
Expected	P2 gets 403/404 without details; first eligible cancellation releases one seat with refund and audit; repeated/late attempts cause no additional inventory change.
Traceability	FR-06, FR-07, FR-09; SEC-01

TC-07 — Administrator Inventory, Authorization and Audit

Preconditions	A1 administrator and P1 passenger exist.
Steps	P1 attempts admin create; A1 creates, edits and disables a trip; inspect state/audit.
Expected	P1 gets 403; A1 changes persist; every change audit has actor, time, action and outcome.
Traceability	FR-08, FR-09; SEC-01

TC-08 — Concurrent Final Seat

Preconditions	Exactly one available seat; 20 authenticated passengers; simulator succeeds.
Steps	Release 20 requests simultaneously from a barrier; inspect bookings and availability.
Expected	At most one confirmation; others report unavailable; final count is zero and never negative.
Traceability	FR-04, FR-05; NFR-03

TC-09 — Transport, Session and Restart Recovery

Preconditions	HTTPS deployment; P1 session; committed booking.
Steps	Inspect transport; sign out/replay token; wait 30 minutes idle and reuse session; restart app; query health/booking.
Expected	Authenticated traffic is HTTPS; signed-out/expired tokens fail; booking persists; health responds within 60 seconds; logs omit passwords/tokens.
Traceability	NFR-04; SEC-02, SEC-03

TC-10 — Performance, Accessibility and Compatibility

Preconditions	10,000 trips; 50 users; current Chrome/Edge/Firefox; 375px/1280px.
Steps	Run 200 searches and 100 bookings; complete required workflows by keyboard at both widths in every browser; inspect accessible names.
Expected	≥95% searches ≤2s and bookings ≤3s under stated conditions; keyboard works; controls have accessible names; primary forms do not scroll horizontally.
Traceability	NFR-01, NFR-02, NFR-05, NFR-06

3. Coverage Summary

Requirement set	Cases
FR-01–FR-03	TC-01–TC-03
FR-04–FR-05	TC-03–TC-05, TC-08
FR-06–FR-07	TC-04, TC-06
FR-08–FR-09	TC-02, TC-04, TC-06, TC-07
NFR-01–NFR-02	TC-10
NFR-03–NFR-04	TC-05, TC-08, TC-09
NFR-05–NFR-06	TC-10
SEC-01–SEC-04	TC-01, TC-02, TC-06, TC-07, TC-09

All 9 functional, 6 non-functional and 4 security requirements have planned verification within ten cases.

4. Execution Record Requirements

Record	Required content
Execution identity	Case ID, build/commit, environment, tester and date/time.
Outcome	Actual result, pass/fail/blocked status and defect ID if applicable.
Evidence	Redacted response/screenshot/log/report references sufficient to reproduce the conclusion.
Review	Retest result after fixes and final requirement coverage sign-off.